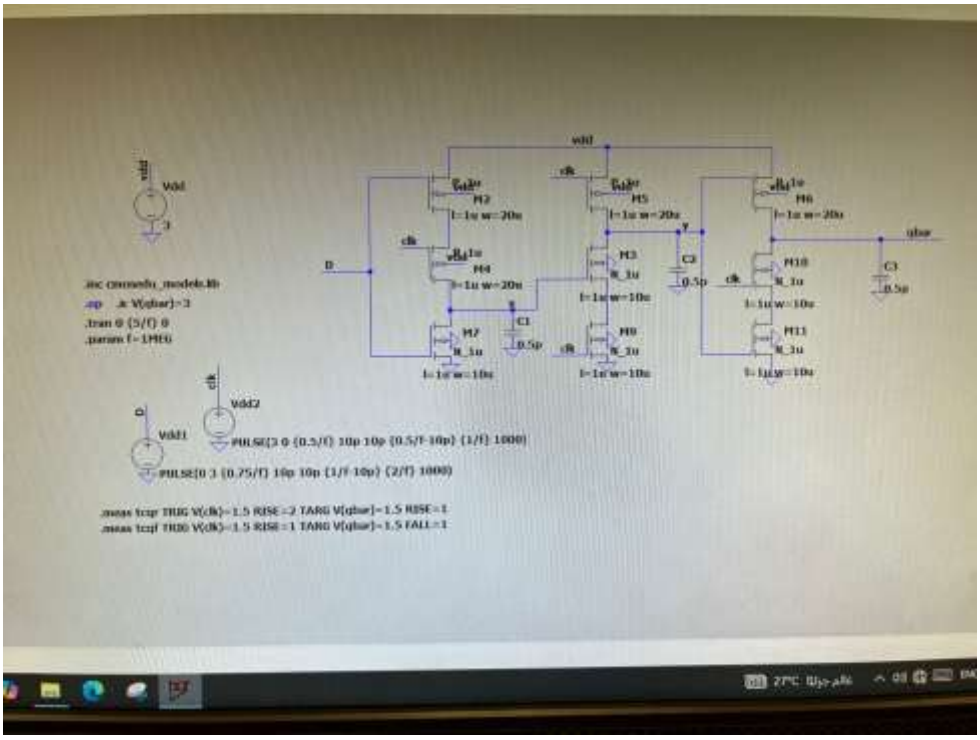
# Lab 4 Digital Circuits

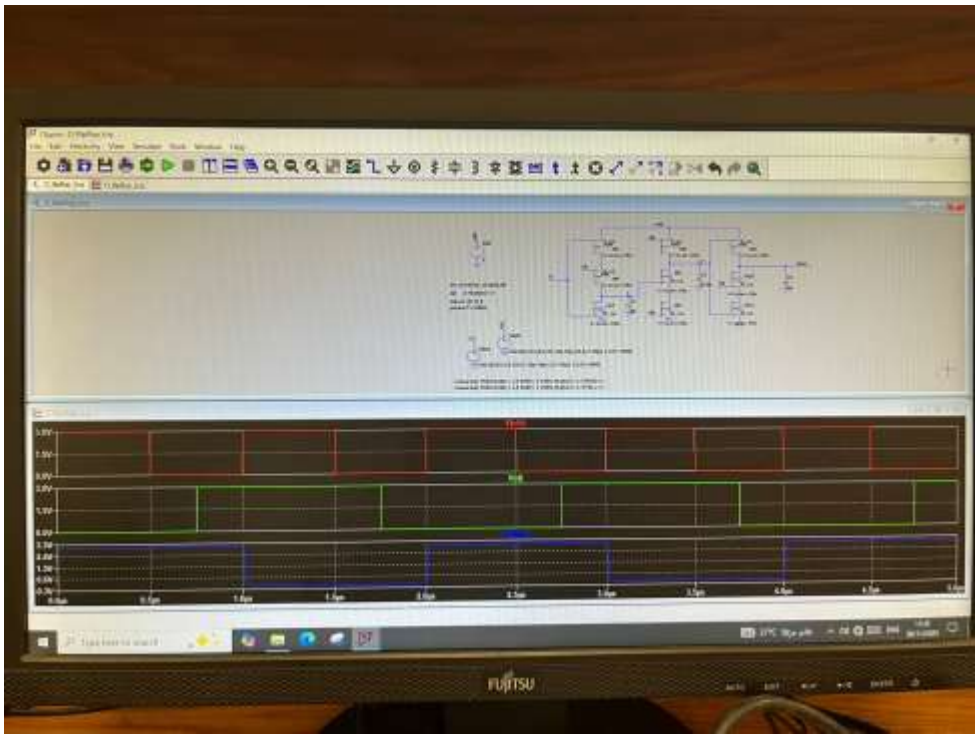
| Name | Section | Attendance slot | Group No. |
|---|---|---|---|
| Amr Hatem Sabry Elmancy | 1 | Sunday ( from 10:00 AM to 12:00 PM ) second rotation | 14 |

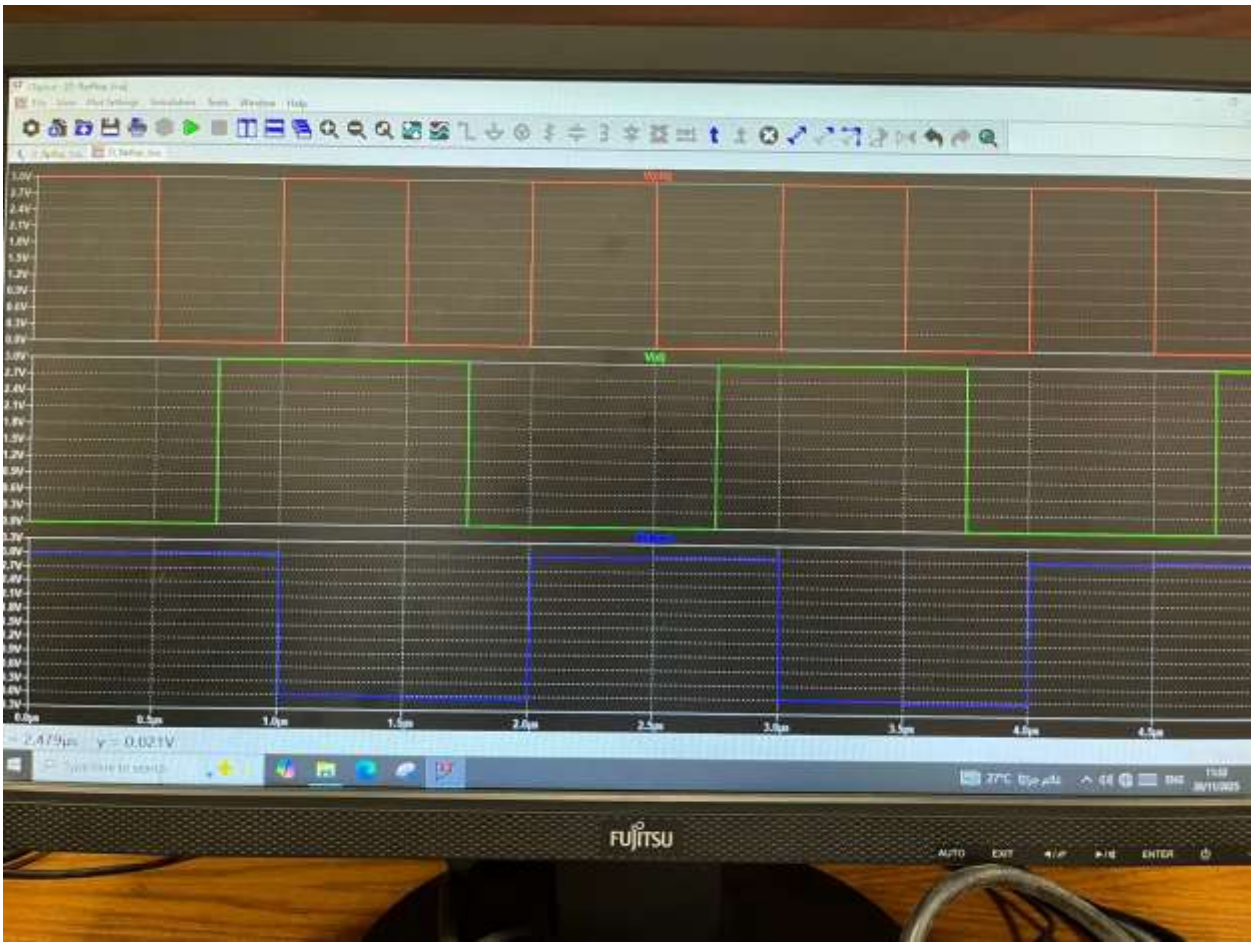
| Name | Section | Attendance slot | Group No. |
|---|---|---|---|
| Abdelrahman Sherif Medhat | 1 | Sunday ( from 10:00 AM to 12:00 PM ) second rotation | 14 |

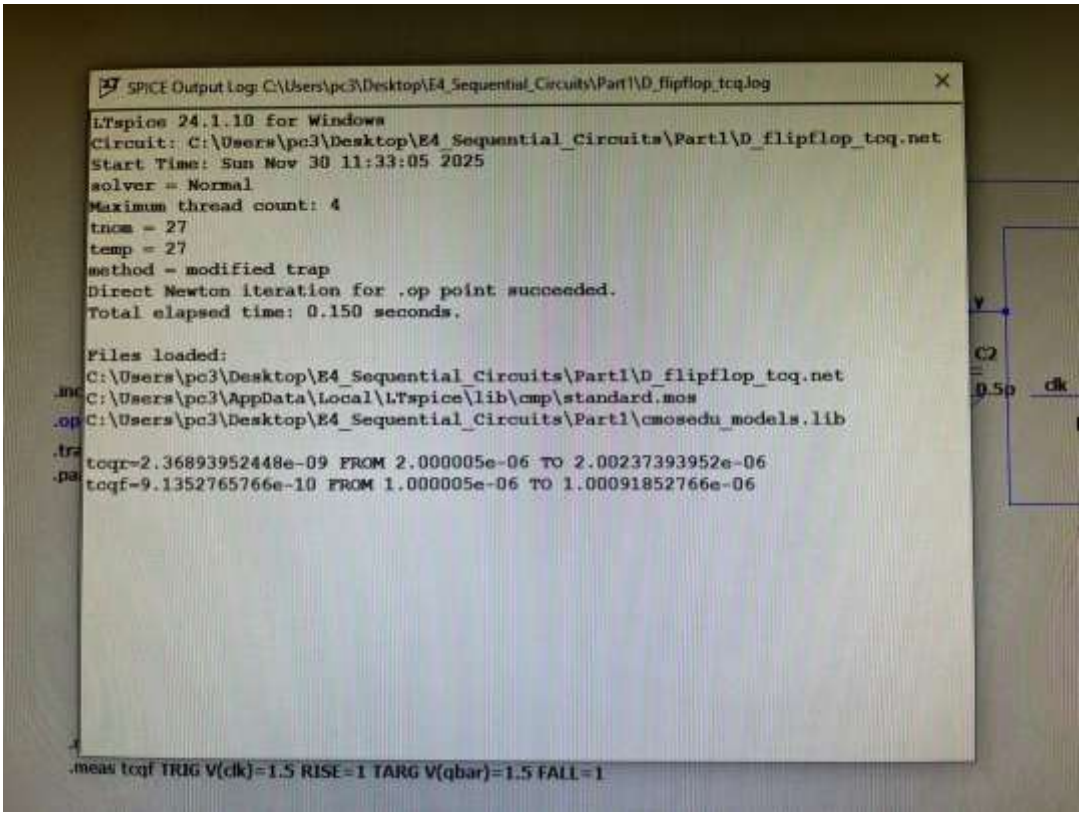
## Part 1: Tc→q delay

The waveforms in different plot planes at the Nets “D, clk, x, y, qbar”.

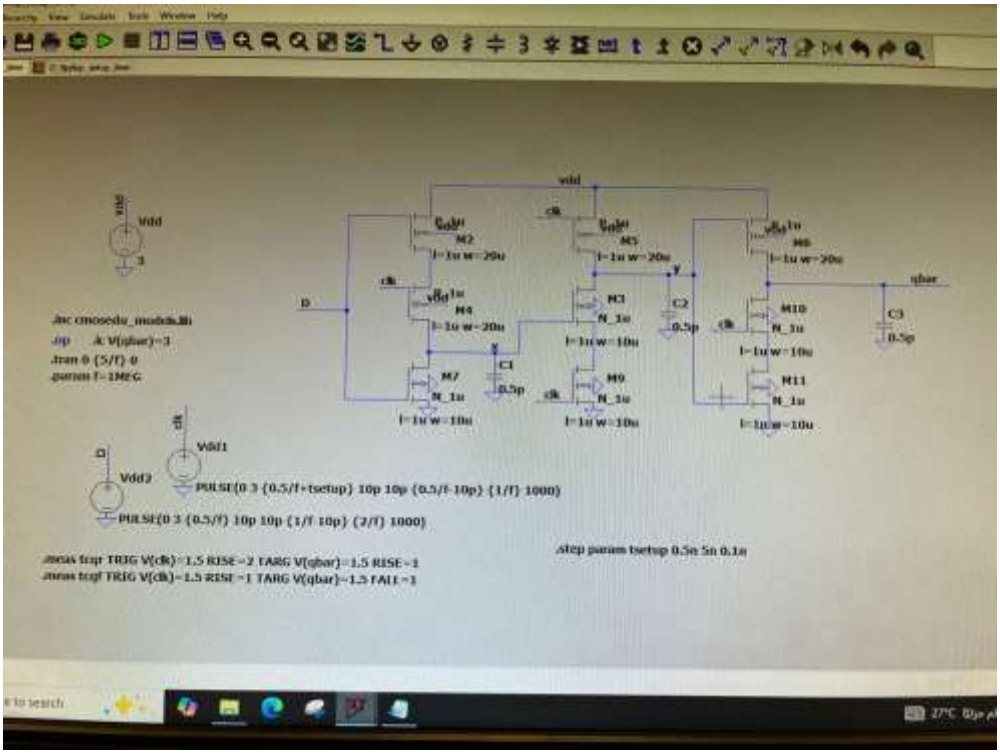


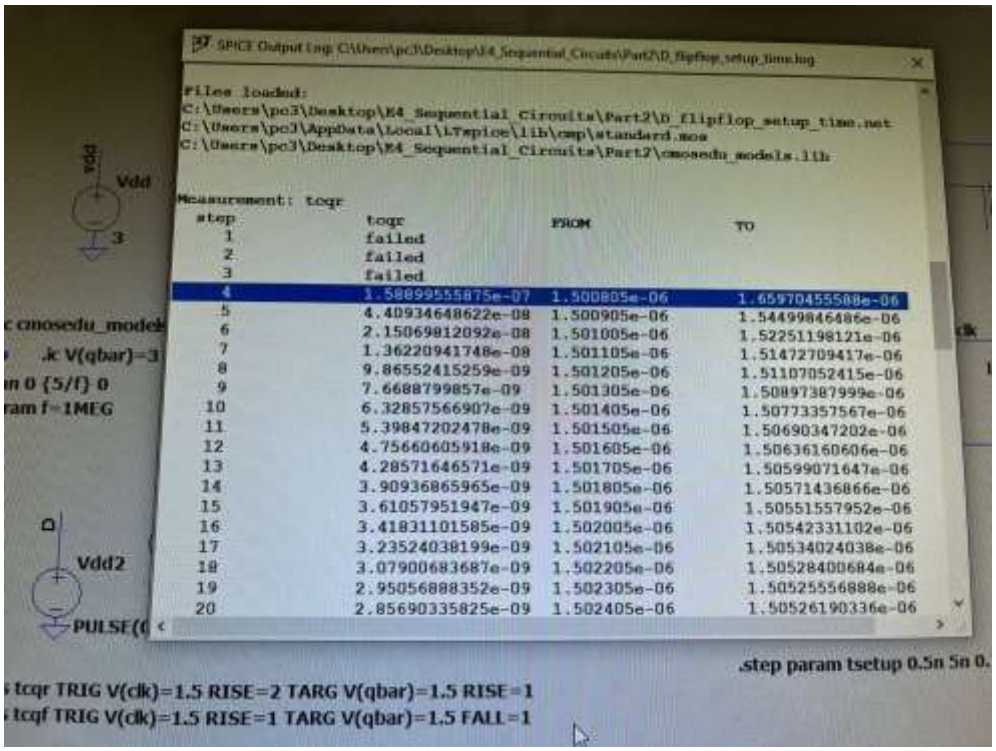



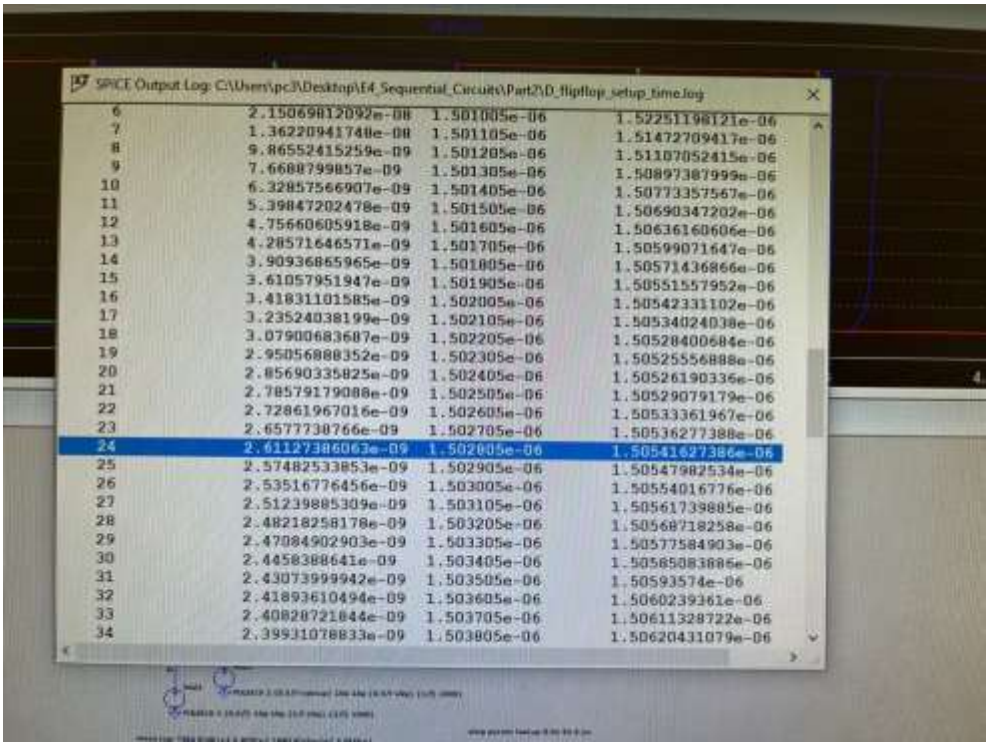

FUJITSU


SPICE Output Log: C:\Users\pc3\Desktop\E4_Sequential_Circuits\Part1\D_flipflop_tcq.log
LTspice 24.1.10 for Windows
Circuit: C:\Users\pc3\Desktop\E4_Sequential_Circuits\Part1\D_flipflop_tcq.net
Start Time: Sun Nov 30 11:33:05 2025
solver = Normal
Maximum thread count: 4
tnom = 27
temp = 27
method = modified trap
Direct Newton iteration for .op point succeeded.
Total elapsed time: 0.150 seconds.
Files loaded:
C:\Users\pc3\Desktop\E4_Sequential_Circuits\Part1\D_flipflop_tcq.net
C:\Users\pc3\AppData\Local\LTspice\lib\cmp\standard.mos
C:\Users\pc3\Desktop\E4_Sequential_Circuits\Part1\cmosedu_models.lib
tcqr=2.36893952448e-09 FROM 2.000005e-06 TO 2.00237393952e-06
tcqf=9.1352765766e-10 FROM 1.000005e-06 TO 1.00091852766e-06
C2
0.5p
clk
.meas tcqf TRIG V(clk)=1.5 RISE=1 TARG V(qbar)=1.5 FALL=1

# Part 2: Setup time






| | | | |
|---|---|---|---|
| 6 | 2.15069812092e-08 | 1.501005e-06 | 1.52251198121e-06 |
| 7 | 1.36220941748e-08 | 1.501105e-06 | 1.51472709417e-06 |
| 8 | 9.86552415259e-09 | 1.501205e-06 | 1.51107052415e-06 |
| 9 | 7.66807998857e-09 | 1.501305e-06 | 1.50897307999e-06 |
| 10 | 6.32857566907e-09 | 1.501405e-06 | 1.50773357567e-06 |
| 11 | 5.39847202478e-09 | 1.501505e-06 | 1.50690347202e-06 |
| 12 | 4.75660605918e-09 | 1.501605e-06 | 1.50636160606e-06 |
| 13 | 4.28571646571e-09 | 1.501705e-06 | 1.50599071647e-06 |
| 14 | 3.90936865965e-09 | 1.501805e-06 | 1.50571436866e-06 |
| 15 | 3.61057951947e-09 | 1.501905e-06 | 1.50551557952e-06 |
| 16 | 3.41831101585e-09 | 1.502005e-06 | 1.50542331102e-06 |
| 17 | 3.23524038199e-09 | 1.502105e-06 | 1.50534024038e-06 |
| 18 | 3.07900683687e-09 | 1.502205e-06 | 1.50528400684e-06 |
| 19 | 2.95056888352e-09 | 1.502305e-06 | 1.50525556888e-06 |
| 20 | 2.85690335825e-09 | 1.502405e-06 | 1.50526190336e-06 |
| 21 | 2.78579179088e-09 | 1.502505e-06 | 1.50529079179e-06 |
| 22 | 2.72861967016e-09 | 1.502605e-06 | 1.50533361967e-06 |
| 23 | 2.6577738766e-09 | 1.502705e-06 | 1.50536277388e-06 |
| 24 | 2.61127386063e-09 | 1.502805e-06 | 1.50541627386e-06 |
| 25 | 2.57482533853e-09 | 1.502905e-06 | 1.50547982534e-06 |
| 26 | 2.53516776456e-09 | 1.503005e-06 | 1.5055401677 6e-06 |
| 27 | 2.51239885309e-09 | 1.503105e-06 | 1.50561739885e-06 |
| 28 | 2.48218258178e-09 | 1.503205e-06 | 1.50568718258e-06 |
| 29 | 2.47084902903e-09 | 1.503305e-06 | 1.50577584903e-06 |
| 30 | 2.4458388641e-09 | 1.503405e-06 | 1.50585083885e-06 |
| 31 | 2.43073999942e-09 | 1.503505e-06 | 1.5059357 4e-06 |
| 32 | 2.41893610494e-09 | 1.503605e-06 | 1.5060239361e-06 |
| 33 | 2.40820721844e-09 | 1.503705e-06 | 1.50611328722e-06 |
| 34 | 2.39931078833e-09 | 1.503805e-06 | 1.50620431079e-06 |

**Measure worst case Tcq:**

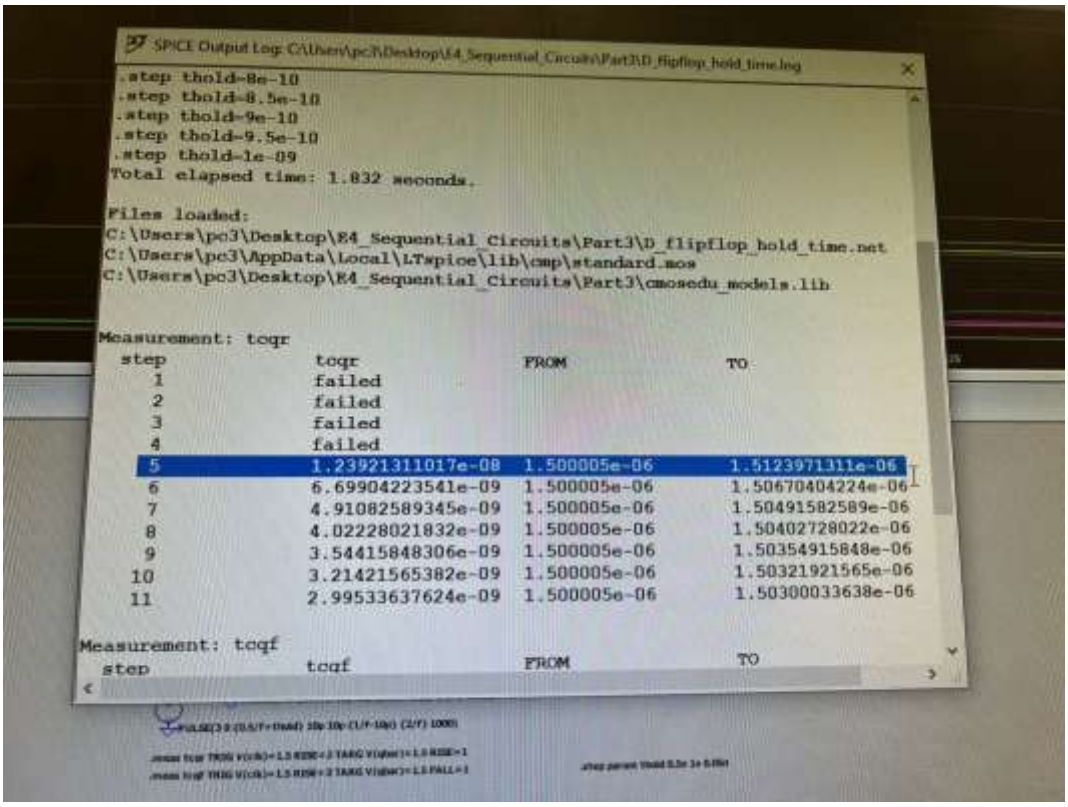
**At tsetup 0.9n**

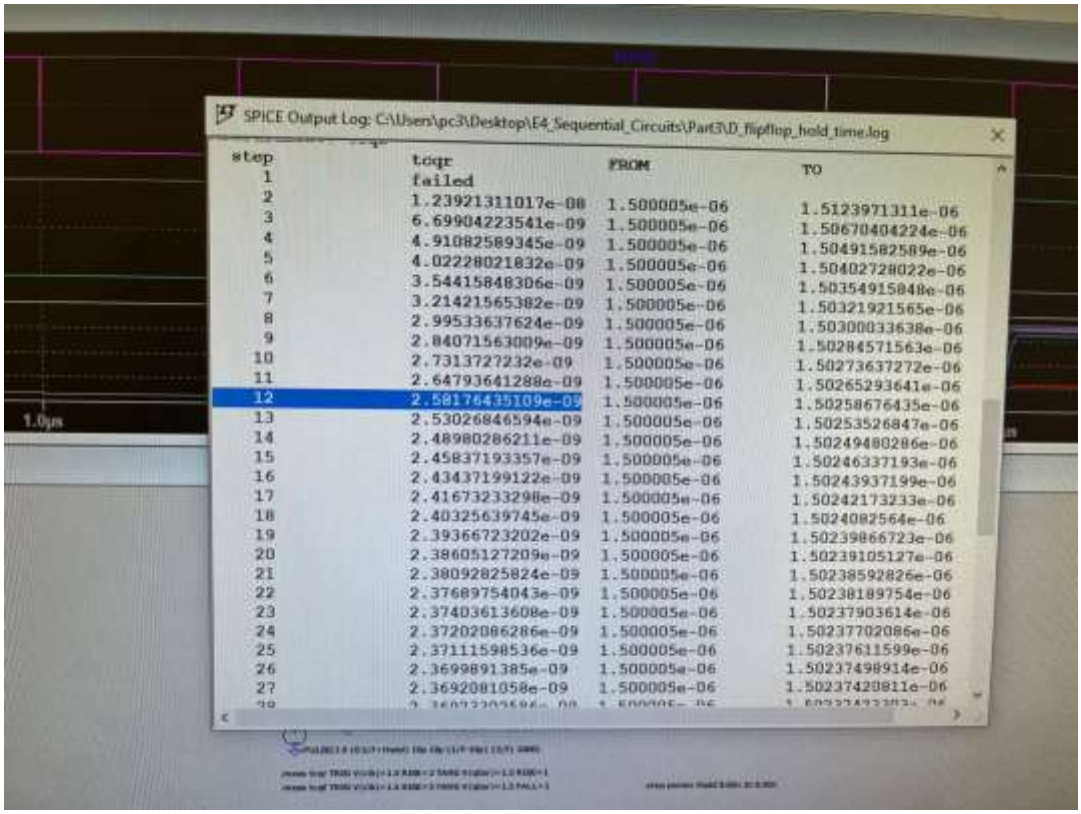
**Tcq=158.9nanosecond**

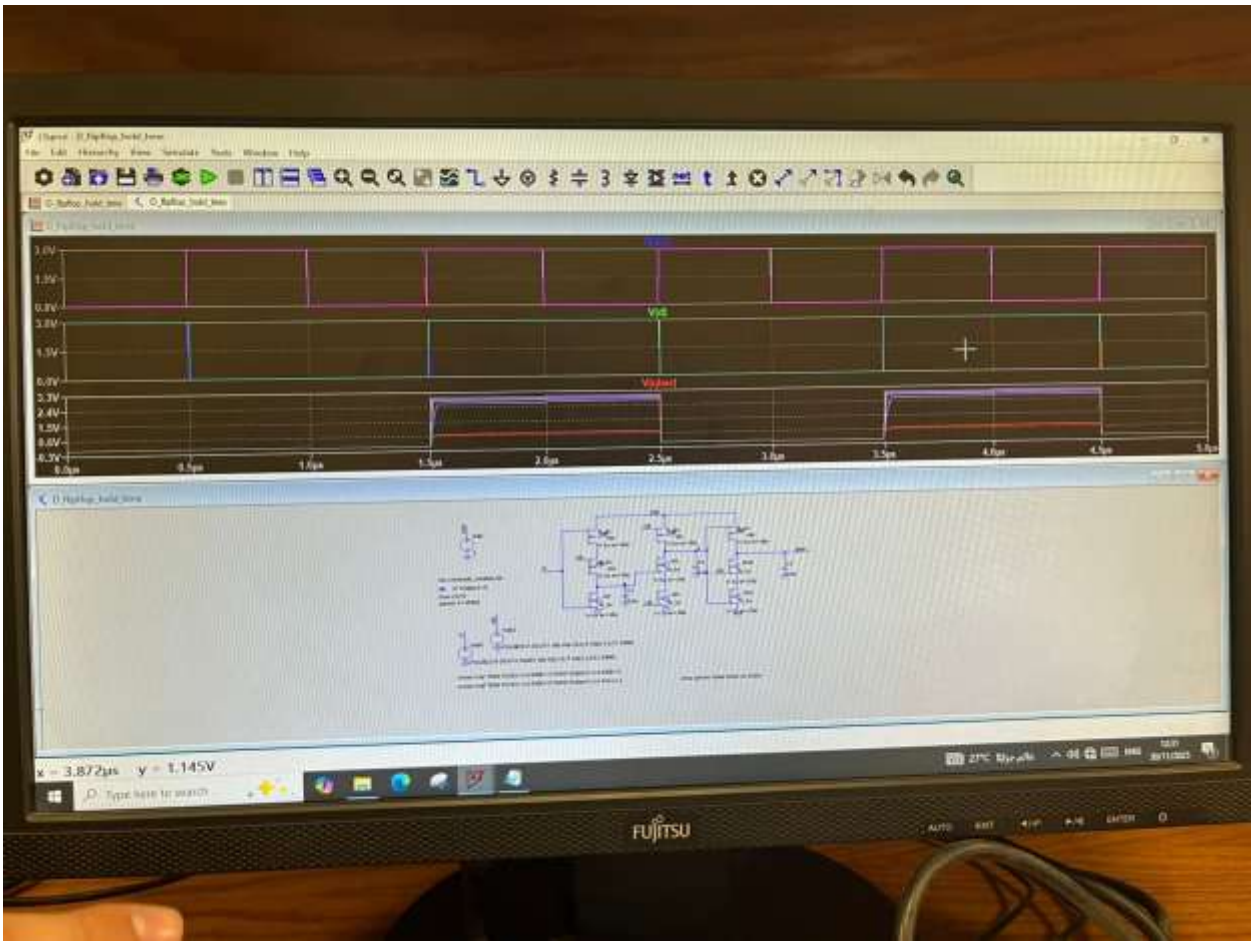
**Get Tsetup at which Tcq increased by 10%:**

**Tsetup=2.9 nanosecond**

# Part 3: hold time






**Measure worst case Tcq:**

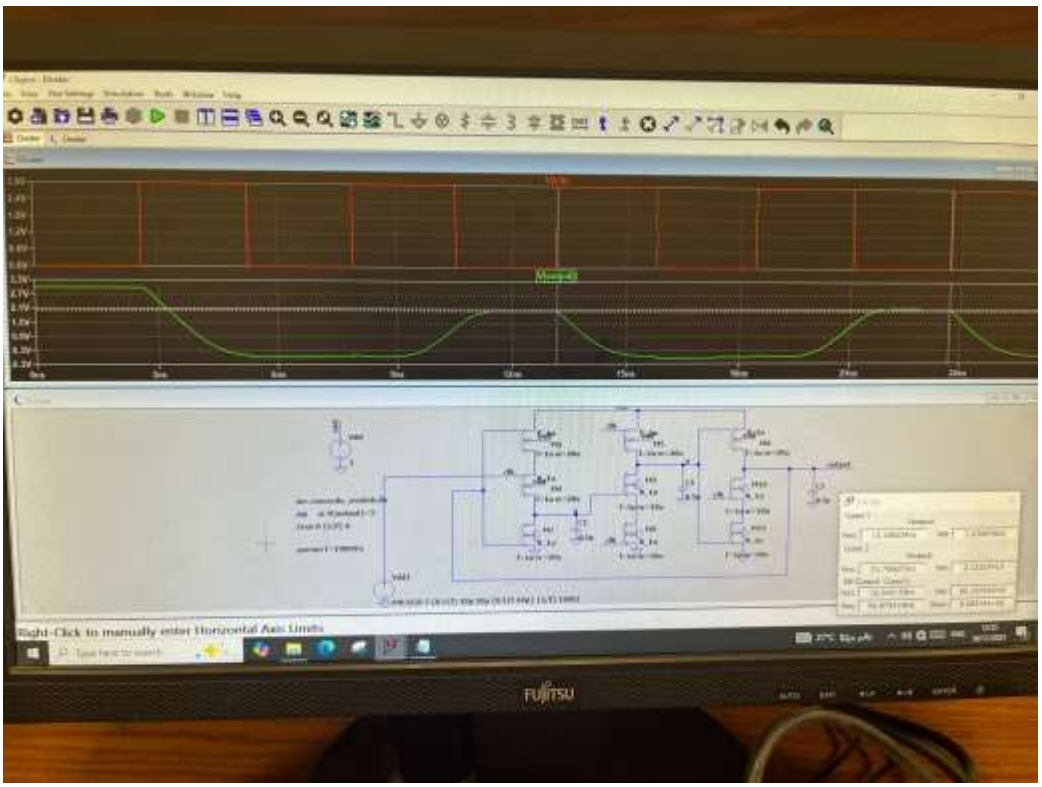
**At thold 0.9n**

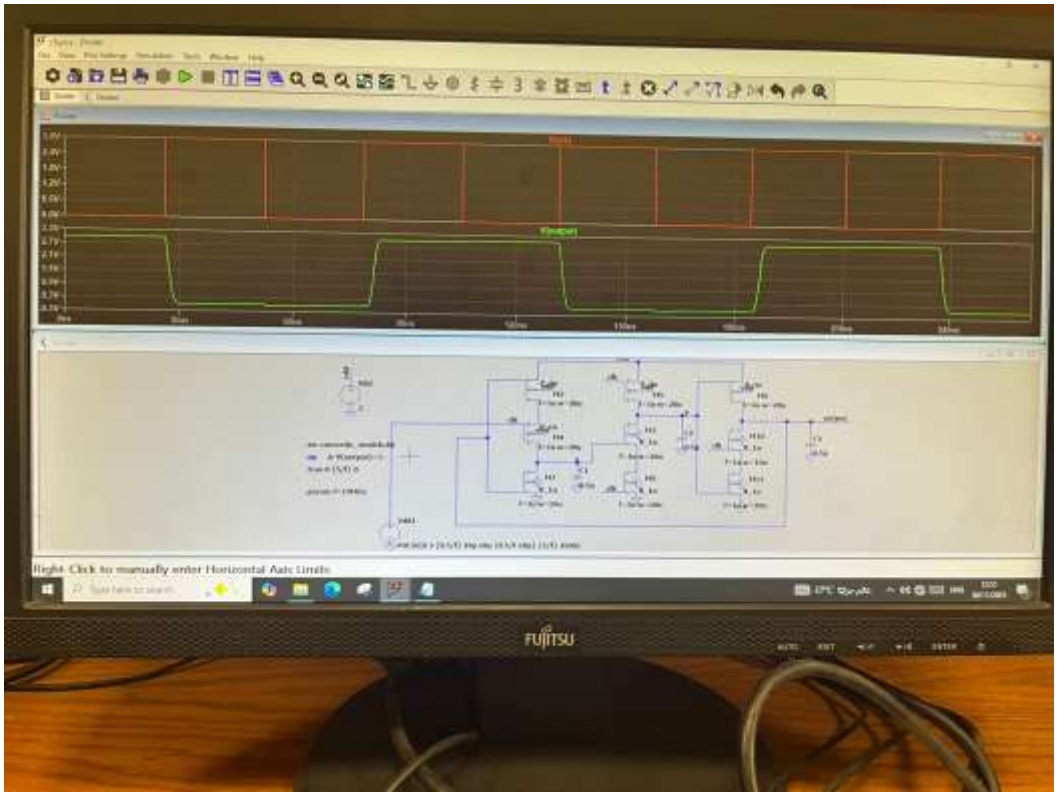
**Tcq=12.39 nanosecond**

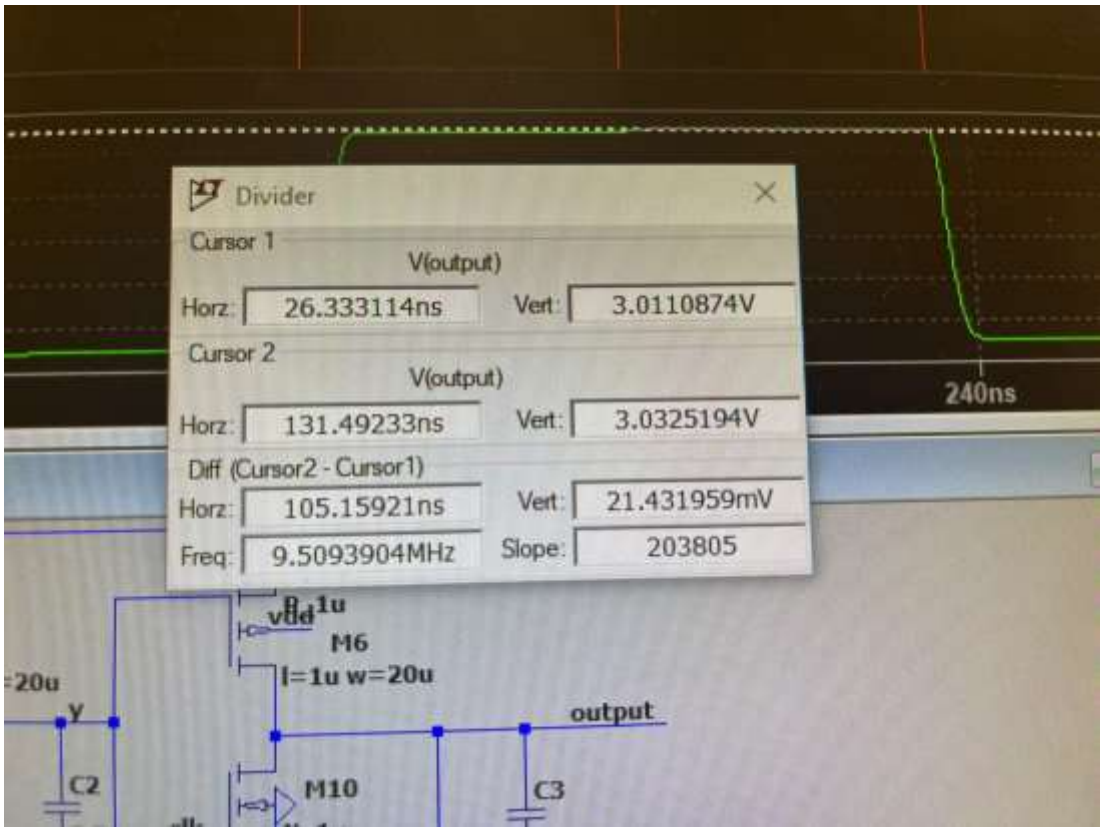
**Get Thold at which Tcq increased by 10%:**

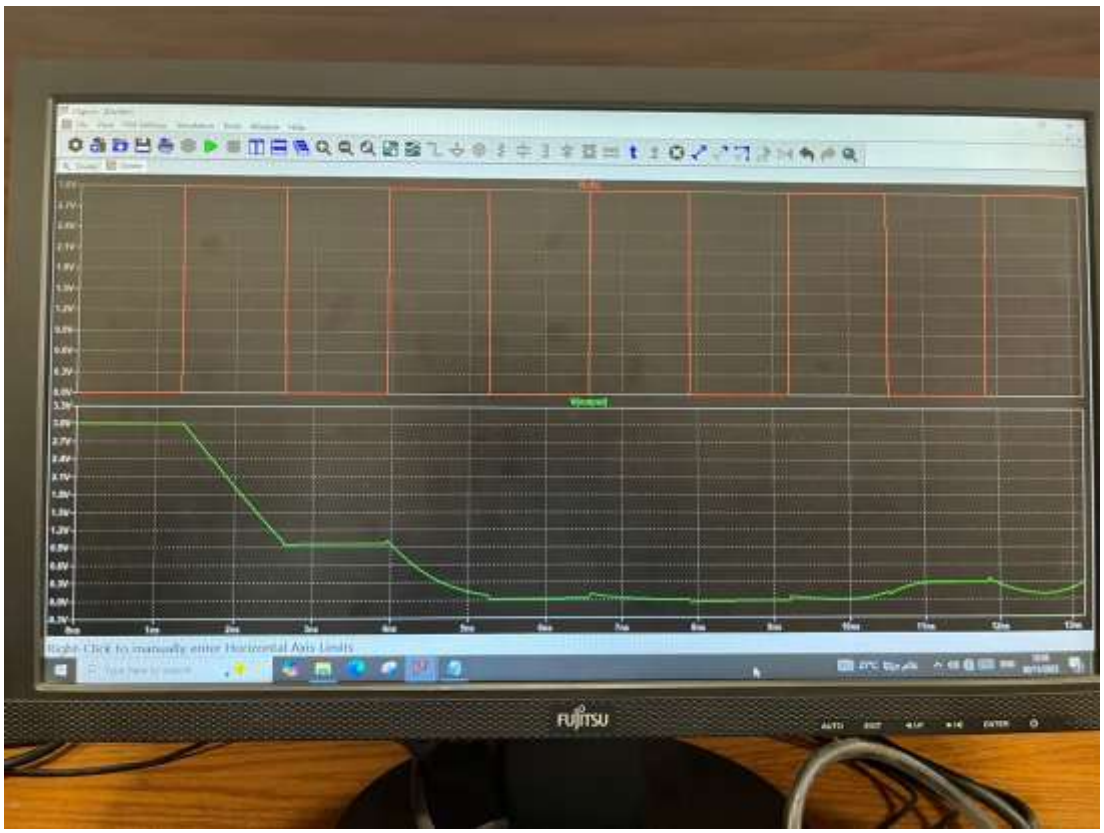
**Thold=1.25 nanosecond**

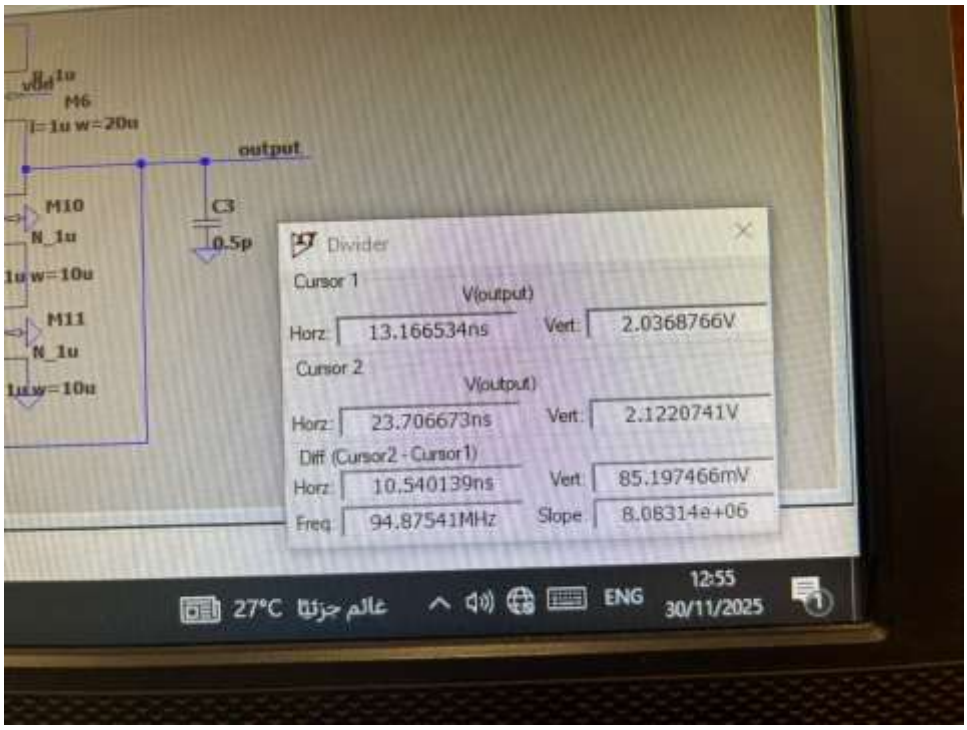
# Part 3: hold time

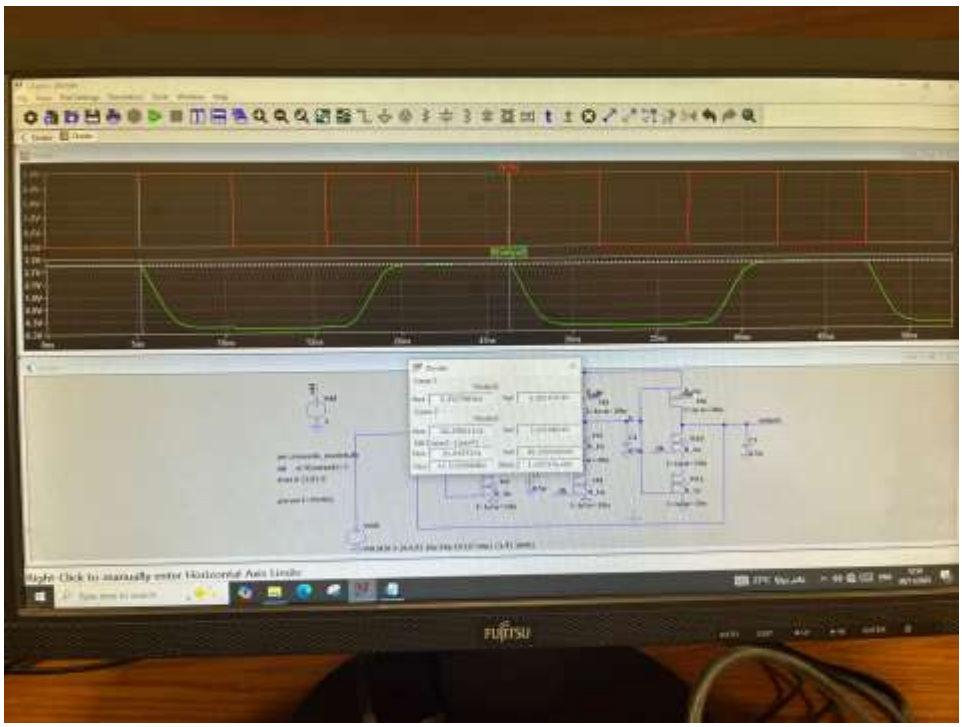



Divider
Cursor 1
V(output)
Horz: 26.333114ns
Vert: 3.0110874V
Cursor 2
V(output)
Horz: 131.49233ns
Vert: 3.0325194V
Diff (Cursor2 - Cursor1)
Horz: 105.15921ns
Vert: 21.431959mV
Freq: 9.5093904MHz
Slope: 203805
240ns
vdd
1u
M6
l=1u w=20u
:20u
y
output
C2
M10
C3


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**Calculate and simulate fmax:**

**Fmax=189.8Mhz**